

What is Neurosymbolic Computing?

Definition: Neurosymbolic Computing integrates neural network–based learning (subsymbolic) with formal symbolic reasoning, combining the pattern-recognition strength of deep learning with the interpretability, compositionality, and logical rigor of classical AI.

Neural Learning

Learns from data;
handles perception,
language & uncertainty

Symbolic Reasoning

Logic rules, formal
proofs, structured
knowledge & planning

Hybrid Integration

LLMs generate code;
solvers verify; rules
guide generation

Why Neurosymbolic AI?

⚠ LLMs Fail at Reasoning

LLMs hallucinate, cannot plan reliably, and produce unfaithful chain-of-thought explanations — pure neural approaches hit a ceiling.

✓ Symbolic Systems Lack Flexibility

Pure logic systems require hand-crafted rules and struggle with ambiguous, noisy natural language — they need neural grounding.

Seminar Structure

14 students · 30 min presentations · 5 thematic topics

1

Motivation for Neurosymbolic Research

Why pure LLMs fall short: unfaithful reasoning, symbolic limitations in math & planning

Introductory

S1 · S2 · S3

2

Neurosymbolic Code Generation for QA

PAL, Logic-LM, Prolog-based reasoning, causal proofs, NL→FOL translation

Core

S4 · S5 · S6 · S7 · S8

3

Knowledge-Grounding for NeSy QA Systems

ProSLM, RuleRAG, RuAG, theorem proving (LeanDojo), adaptive retrieval (SymRAG)

Core

S9 · S10 · S11 · S12 · S13

4

Neurosymbolic Applications

Medicine, clinical decision support, legal compliance & financial analysis

Applied

S14 · S15 · S16 · S17

5

Neurosymbolic Data Generation for Evaluation

ProofWriter & LogicNLI: synthetic benchmarks for logical reasoning evaluation

Methodological

S18 · S19

Paper Overview — 19 Papers · 14 Student Presentations

Topic 1 Motivation	Topic 2 Code Gen / QA	Topic 3 Knowledge-Grounding	Topic 4 Applications	Topic 5 Data Generation
[S1] Turpin et al. 2023 Unfaithful CoT Explanations	[S4] Gao et al. 2023 PAL	[S9] Vakharia et al. 2024 ProSLM	[S14] Prenosil et al. 2025 Medical Report AI	[S18] Tafjord et al. 2021 ProofWriter
[S2] Mirzadeh et al. 2024 GSM-Symbolic: Math Limits	[S5] Pan et al. 2023 Logic-LM	[S10] Chen et al. 2024 RuleRAG	[S15] Tan et al. 2025 Clinical Decision Support	[S19] Tian et al. 2021 LogicNLI
[S3] Kambhampati et al. 2024 LLM-Modulo Planning	[S6] Di et al. 2025 LoRP (Prolog)	[S11] Zhang et al. 2024 RuAG	[S16] Hsia et al. 2025 Financial Legal Analysis	
	[S7] Yang et al. 2025 Neuro-Symbolic Proofs	[S12] Yang et al. 2023 LeanDojo	[S17] Kant et al. 2025 Legal Reasoning	
	[S8] Putra et al. 2026 NL2Logic	[S13] Hakim et al. 2025 SymRAG		

What Students Will Learn

01 Understand the Limits of Pure LLMs

Critically evaluate where and why large language models fail — including hallucination, unfaithful reasoning chains, and inability to plan — motivating the neurosymbolic paradigm.

02 Master Neurosymbolic Architectures

Understand how LLMs are combined with symbolic solvers (Prolog, FOL, SMT solvers) for faithful, verifiable QA — including code generation (PAL, Logic-LM) and rule-driven inference.

03 Analyze Knowledge-Grounding Techniques

Compare retrieval-augmented generation approaches enhanced with logical rules (RuleRAG, SymRAG) and formal theorem-proving environments (LeanDojo).

04 Evaluate Real-World Applications

Assess neurosymbolic AI in high-stakes domains (medicine, law, finance) and evaluate trade-offs between explainability, reliability, and scalability.

05 Engage with Reasoning Benchmarks

Understand how synthetic datasets (ProofWriter, LogicNLI) are constructed for evaluating first-order logical reasoning and critically assess benchmark design choices.

06 Develop Scientific Presentation Skills

Present a research paper to peers with clarity and depth — summarizing contributions, methodology, results, and open challenges in a 30-minute talk with slides.